

NATIONAL UNIVERSITY OF SCIENCES AND TECHNOLOGY

School of Electrical Engineering and Computer Sciences



Digital Dice

A Digital Logic Design Project using 555 Timer, 4017 Decade Counter & 74LS47 BCD Decoder

COURSE NAME: Computer Architecture and Logic Design

Project Report

SUBMITTED BY	Group Members
1.	Alisha Maryam Habib (555847)
2.	Bismah Adil (551853)
3.	Meerab Eman (575014)
4.	Uswa Fatima (575207)
CLASS	BESE-16-A
LAB ENGINEER	Ma'am Rafia Ahmad

INTRODUCTION

Project Summary

This project presents the design and implementation of a Digital Dice using fundamental digital logic components, without any microcontroller. The system simulates the roll of a standard six-sided die by generating a pseudo-random number between 1 and 6 and displaying it on a 7-segment display. The circuit leverages the oscillating capability of a 555 Timer IC, the counting capability of a 4017 Decade Counter, and the display decoding capability of a 74LS47 BCD-to-7-Segment Decoder.

When the push button is pressed and released, the 555 timer generates rapid clock pulses that drive the 4017 counter through states 0 to 5. Upon release, the counter freezes at a pseudo-random state corresponding to a die face (1 through 6), which is decoded and shown on the 7-segment display. The result appears random to the human eye due to the high clock frequency relative to human reaction time.

Project Outcomes

The primary outcomes of this project are:

- Design and implement a Digital Dice using only discrete digital logic ICs.
- Generate a pseudo-random number (1-6) using a 555 timer oscillator and a decade counter.
- Display the result on a common-anode 7-segment display via a 74LS47 BCD decoder.
- Achieve a fully microcontroller-free design that is cost-effective and easy to replicate.

Project Scope

- Focus on digital logic-based random number simulation.
- Use of 555 Timer in astable mode as a clock source.
- Utilize 4017 Decade Counter for counting from 0-5 (mapped to 1-6).
- BCD-to-7-Segment decoding using 74LS47 IC.
- Provide a single push button interface for intuitive dice-roll operation.
- Provide a foundation for future enhancements such as multiple dice, score tracking, or wireless display.

Design and Implementation

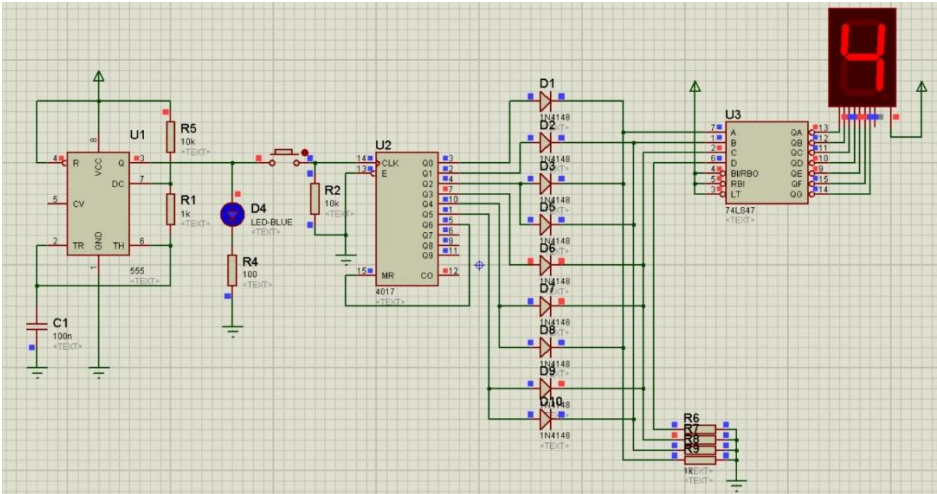
Detailed Explanation of Hardware Used

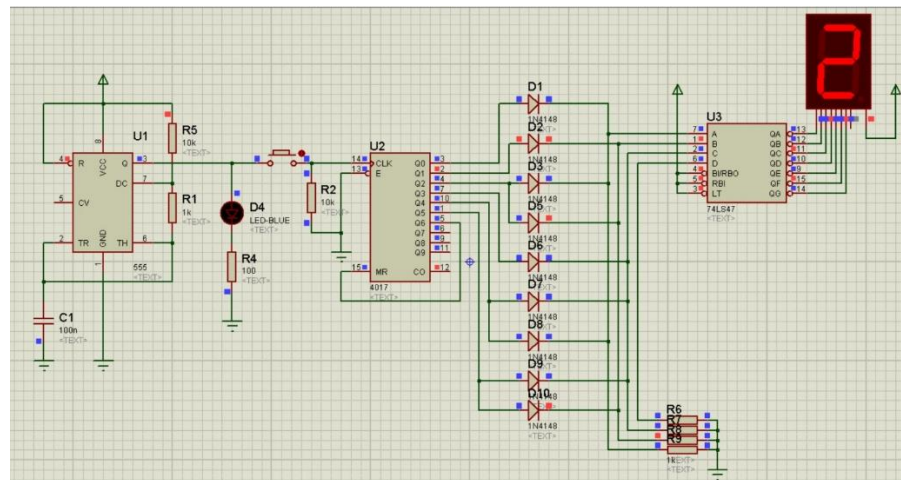
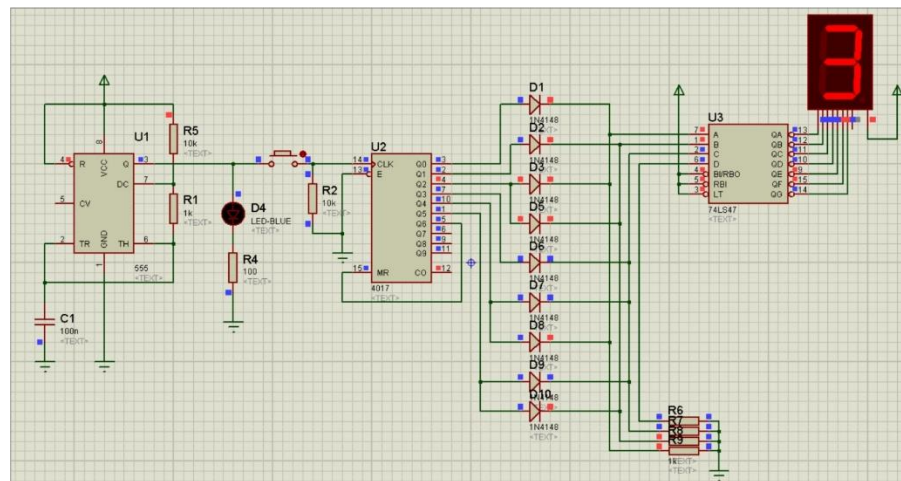
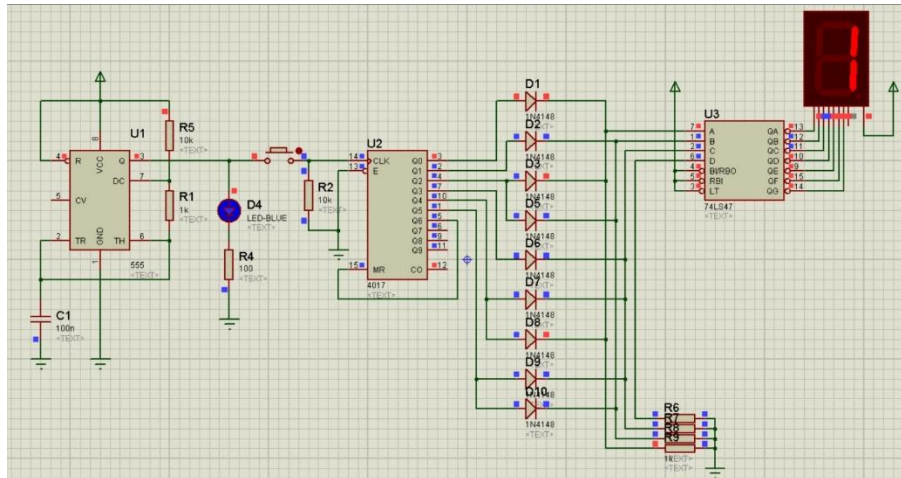
The Digital Dice system comprises the following key components:

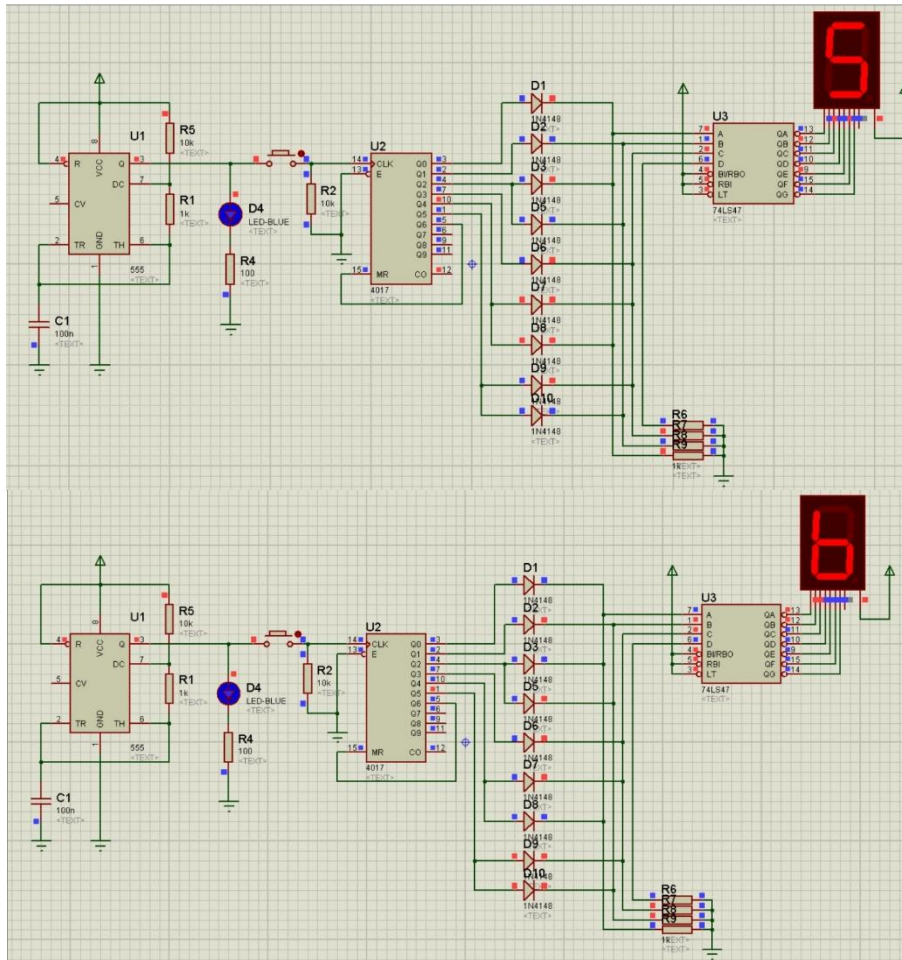
#	Component	Purpose
1	555 Timer IC	Generates continuous clock pulses to drive the counter
2	4017 Decade Counter	Counts clock pulses; outputs Q0-Q5 for dice faces 1-6
3	74LS47 BCD-to-7-Seg Decoder	Decodes BCD input from counter to 7-segment display signals
4	7-Segment Display	Displays the resulting dice number (1 through 6)
5	Push Button	User input to initiate and stop the dice roll
6	1N4148 Diodes (x9)	Implement diode logic to map counter outputs to BCD inputs
7	Red LED	Visual indicator that the circuit is powered and active
8	Resistors (100R)	Current limiting for 7-segment display segments
9	Resistors (1k x6)	Pull-down/current limiting for counter and logic lines
10	Resistors (10k x2)	Timing resistors for 555 timer astable configuration
11	Capacitor (100uF)	Timing capacitor for 555 timer frequency control
12	Capacitor (104pF)	Decoupling/noise bypass capacitor
13	9V Power Supply	Powers the entire circuit
14	Solderless Breadboard	Circuit assembly and prototyping platform

Circuit Diagram

The circuit schematic, simulated in Proteus, is provided in the appendix of this report. The diagram illustrates all interconnections between the 555 Timer, 4017 Decade Counter, diode logic network, 74LS47 BCD decoder, and the 7-segment display.







Truth Table

The following truth table represents the mapping of the 4017 counter output states to BCD inputs of the 74LS47 decoder, and the resulting digit displayed on the 7-segment display:

Counter State	Q0 (D)	Q1 (C)	Q2 (B)	Q3 (A)	BCD (DCBA)	Display
Q0 Active	1	0	0	0	0001	1
Q1 Active	0	1	0	0	0010	2
Q2 Active	0	0	1	0	0011	3
Q3 Active	0	0	0	1	0100	4
Q4 Active	1	0	0	1	0101	5
Q5 Active	0	1	1	0	0110	6
Q6 (Reset)	-	-	-	-	-	Reset to Q0

Inputs and Outputs

1. Inputs to the System

The Digital Dice accepts a single primary input:

Push Button (Roll Control)

- Function: Initiates and halts the dice-rolling sequence.
- Logic when HELD (Pressed): The 555 timer is enabled and continuously clocks the 4017 counter at high speed. The 7-segment display cycles rapidly through digits 1-6, appearing as a blur to the eye.
- Logic when RELEASED: The timer stops clocking, the counter freezes at its current state, and the display holds the final digit — the result of the dice roll.

2. Outputs Generated

7-Segment Display Output

The 7-segment display shows a digit from 1 to 6, determined by the counter state at the moment the push button is released. The 74LS47 decoder converts the BCD signal from the diode logic network into the appropriate segment drive signals (a-g) for the common-anode display.

LED Indicator Output

A red LED remains illuminated as long as power is supplied to the circuit. It serves as a power-on indicator, confirming the circuit is operational before a roll is attempted.

3. Output States Summary

Button State	Display Output	Description
Not Pressed (0)	Last rolled digit	Display holds frozen; circuit is idle
Pressed (1)	Cycling 1 → 6 rapidly	Counter cycles at high speed; display blurs
Released (0 after 1)	Random digit 1-6	Counter freezes; final dice value displayed

Step-by-Step Implementation Process

1. Component Gathering and Verification

- Gather all components: 555 Timer, 4017 Decade Counter, 74LS47, 7-segment display (common anode), 9x 1N4148 diodes, resistors (100R, 1k x6, 10k x2), capacitors (100uF, 104pF), push button, red LED, solderless breadboard, and 9V supply.
- Test each component with a multimeter or component tester to verify functionality before assembly.

2. Clock Generation (555 Timer in Astable Mode)

- Connect the 555 Timer with $R1 = 10k$, $R2 = 10k$, and $C = 100\mu F$ to produce a clock frequency of approximately 4.8 Hz for simulation visibility, or replace C with 104pF for a fast (imperceptible) rolling speed in hardware.
- The output at Pin 3 of the 555 serves as the clock signal for the 4017 counter.
- Connect a 104pF capacitor between Pin 5 and GND for noise decoupling.

3. Counter Configuration (4017 Decade Counter)

- Connect the CLK input (Pin 14) of the 4017 to Pin 3 of the 555 Timer.
- Connect the Clock Enable (Pin 13) to GND to keep the counter always enabled.
- Connect the Reset (Pin 15) to the Q6 output (Pin 5) of the 4017. This ensures the counter automatically resets after state 5 (counting 0-5, representing dice faces 1-6).

4. Diode Logic Network (BCD Encoding)

- Since the 4017 produces one-hot outputs (only one output high at a time), 1N4148 diodes are used to implement OR-gate logic that maps the counter outputs to the correct BCD representation (A, B, C, D) required by the 74LS47 decoder.
- Connect the diode anodes to the appropriate Q outputs and cathodes to the BCD lines, following the mapping in the truth table above. 1k resistors pull BCD lines low as default.

5. BCD-to-7-Segment Decoding (74LS47)

- Connect the BCD inputs A, B, C, D of the 74LS47 to the corresponding diode logic outputs.
- Connect the segment outputs (a-g) through 100R current-limiting resistors to the corresponding segment inputs of the common-anode 7-segment display.
- Connect the common anode pin of the display to VCC (+5V or +9V depending on display rating).
- Tie LT (Lamp Test), RBI (Ripple Blanking Input) high, and BI/RBO to VCC to keep the display active.

6. Push Button Interface

- Wire the push button in series with the 555 Timer reset line (Pin 4) or the clock enable line so that releasing the button stops the clock signal reaching the counter, freezing the display.
- Use a 10k pull-up resistor on the button line to ensure a clean logic high when not pressed.

7. Power LED Indicator

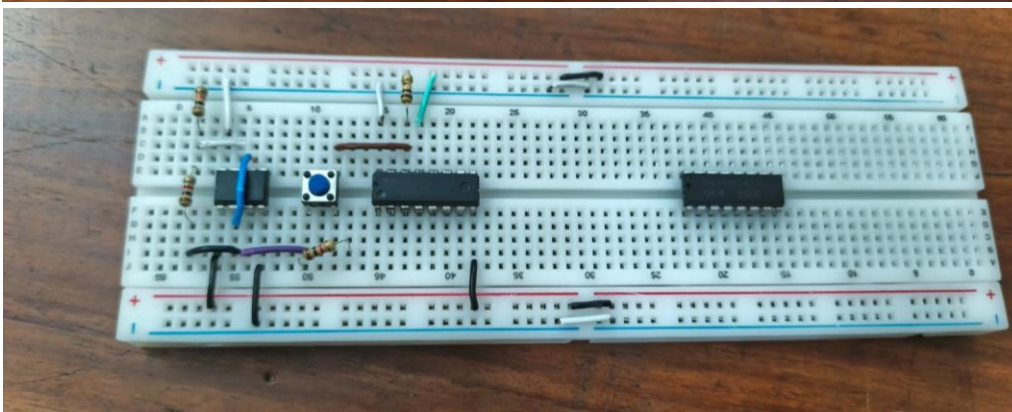
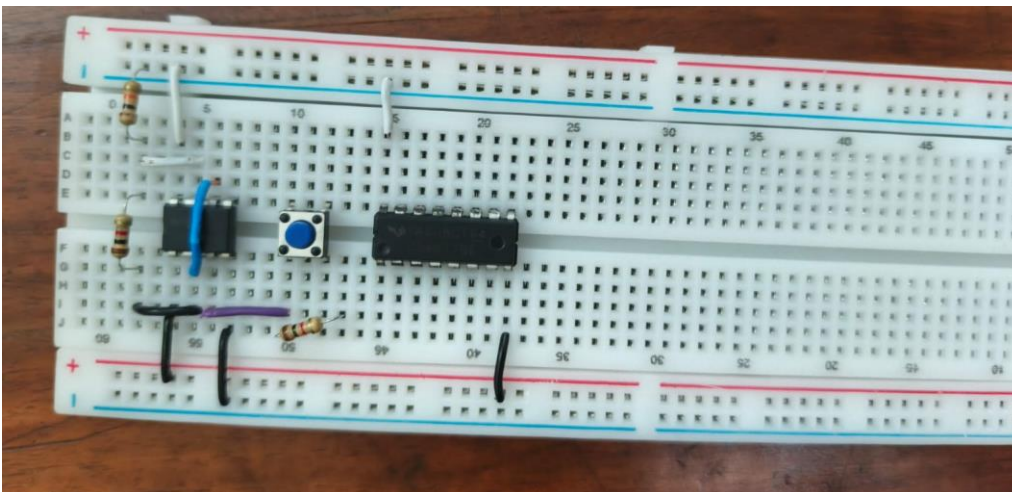
- Connect the red LED in series with a 1k current-limiting resistor between VCC and GND.
- Place it on the breadboard to indicate the circuit is powered on.

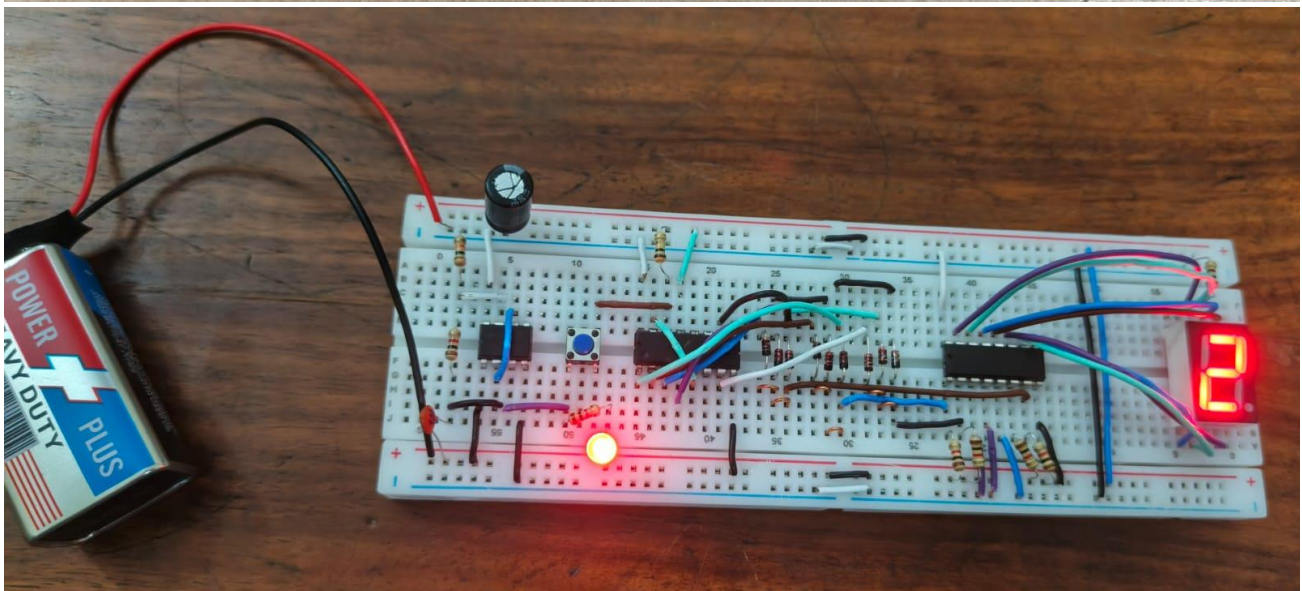
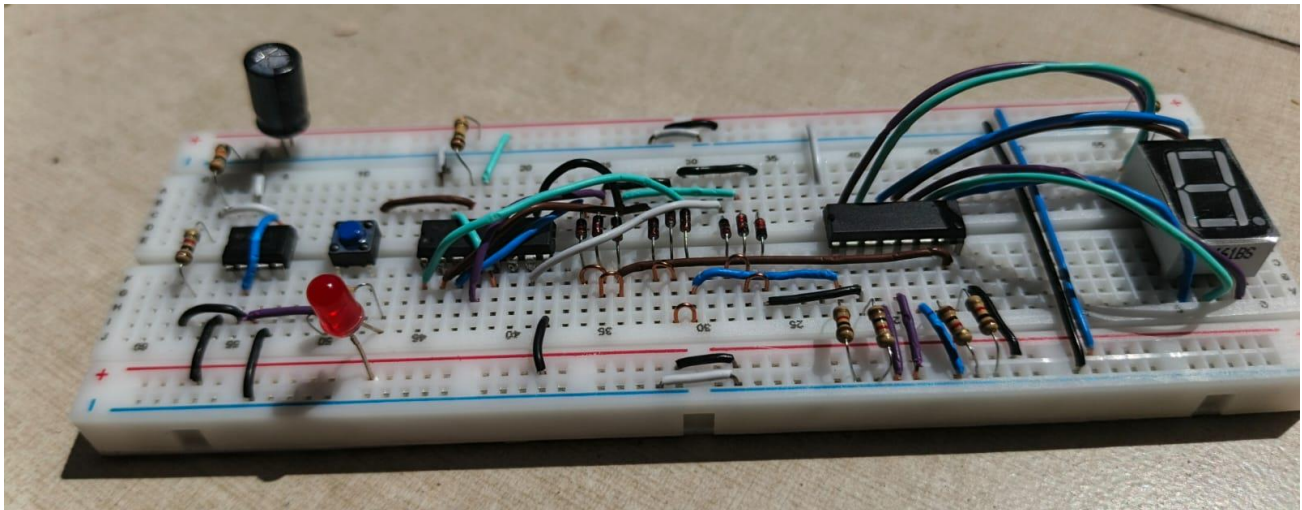
8. Testing and Verification

- Apply 9V power. The LED should illuminate.
- Press and hold the button — the 7-segment display should cycle through 1-6 rapidly.

- Release the button — the display should freeze on a single digit between 1 and 6.
- Repeat multiple times to verify pseudo-randomness and correct digit range.

Step Wise Working:





Results and Analysis

Proteus Simulation Results

The complete circuit was simulated in Proteus Design Suite prior to hardware assembly. The simulation successfully validated the circuit's functionality, including:

- Stable clock signal generation from the 555 Timer at the expected frequency.
- Correct sequential counting of the 4017 from Q0 to Q5, with automatic reset at Q6.
- Accurate BCD encoding via the diode logic network, producing correct BCD codes for digits 1 through 6.
- Correct decoding by the 74LS47 and proper segment activation on the common-anode 7-segment display.
- Clean freeze behavior when the clock enable signal is removed (button released).

Observation: No timing conflicts or logic errors were detected in the final simulation. The circuit schematic is provided in the Appendix.

Experimental Hardware Results

- The dice rolled through all six values cleanly when the button was held.
- Releasing the button reliably froze the display on a pseudo-random value between 1 and 6.
- The display showed no ghosting or incorrect segment activation.
- The circuit operated stably at 9V with no overheating of any component.
- The red LED indicator remained lit consistently throughout testing.

Analysis

Successes:

- Successfully achieved a microcontroller-free random dice simulation using pure digital logic.
- The push-button interface proved highly intuitive — press to roll, release to reveal.
- Proteus simulation results closely matched the physical hardware behavior.
- Diode-based BCD encoding proved a cost-effective alternative to dedicated encoding ICs.

Limitations / Discrepancies:

- Pseudo-randomness: The result is deterministic based on when the button is released. A truly random result requires the use of a ring oscillator or random seed, which was outside the scope of this project.
- Bounce Noise: Without hardware debouncing (e.g., a Schmitt trigger or RC debounce circuit), rapid button presses occasionally caused double-counting.
- Display Brightness: At 9V, some segments appeared slightly dimmer than others due to minor resistor tolerance variations. Calibration with precision resistors would improve uniformity.
- Scalability: Rolling two dice simultaneously or adding a score counter would require significant circuit expansion.

Appendices

Appendix A: Proteus Circuit Simulation

The Proteus simulation screenshot showing the complete circuit schematic and live 7-segment output is provided as a separate image file alongside this report (see: digital_dice_proteus.jpeg).

Appendix B: Component List with Quantities

#	Component	Qty	Notes
1	Solderless Breadboard	1	Full-size
2	555 Timer IC	1	DIP-8 package
3	4017 Decade Counter IC	1	CMOS / TTL
4	74LS47 BCD-to-7-Seg Decoder IC	1	Common Anode driver
5	7-Segment Display (Common Anode)	1	Red, 0.56 inch
6	Push Button	1	Tactile, momentary
7	1N4148 Diode	9	Signal diode
8	Red LED	1	5mm
9	Resistor 100R	1	0.25W
10	Resistor 1k	6	0.25W
11	Resistor 10k	2	0.25W
12	Capacitor 100uF	1	Electrolytic, 16V
13	Capacitor 104pF (0.1uF)	1	Ceramic disc
14	9V Power Supply / Battery	1	DC regulated

Appendix C: Helping Materials and References

1. NE555 / LM555 Timer IC Datasheet – [xx555 Precision Timers datasheet \(Rev. K\)](#)
2. CD4017B CMOS Decade Counter/Divider Datasheet – [CD4017B data sheet, product information and support | TI.com](#)
3. SN74LS47 BCD-to-Seven-Segment Decoder Datasheet – [SN74LS47 data sheet, product information and support | TI.com](#)
4. 1N4148 Signal Diode Datasheet – [1N4148 Datasheet\(PDF\) - Fairchild Semiconductor](#)
5. Driving a 7-Segment Display with a 4017 Counter – [Basic Electronics Tutorials and Revision](#)
6. Astable 555 Timer Calculator – [555 Timer Circuits](#)
7. Proteus Design Suite – [PCB Design and Circuit Simulator Software - Proteus](#)